Road Closure Detection based upon Multi-feature Fusion

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ABSTRACT

Delayed and missed detection of road closure brings a great influence on the quality of the digital map. The existing solutions using trajectory data aim to identify the closed roads according to the drastic drop property in traffic flow. But in actual applications, such methods may lead to the misidentification of a traffic jam as closure, and cannot detect some events like one side closure of two-way road and the closure in the middle of the road. With the occurrence of road closure, there are variations of turning volume of neighboring roads and the increment of U-turn frequency on the closed roads besides the drastic drop of traffic flow. In this paper, we present a high-efficiency road closure detection framework based upon multi-feature fusion, called *RCDM*. It consists of an *off-line road closure feature modeling* part and an *online identification* part. In the off-line phase, we first partition the road network into grids, and then extract road closure features of grids and those of roads from historical data. In the online phase, on the basis of the predictions for road closure features, we screen out closed grid candidates in terms of traffic flow plunge property and further pinpoint the closed road sections according to turning behavior variations of roads. Extensive experimental results on three real data sets from *Chengdu*, *Shanghai* and *Beijing* validate that our method has higher detection accuracy and efficiency compared with the existing methods.

CCS CONCEPTS

• **Information systems → Location based services**.

KEYWORDS

road closure, traffic flow, turning volume, u-turn, grid

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(a) Road closure in the middle of road (b) One side closure of two-way road

Figure 1: Two Special Types of Road Closures

1 INTRODUCTION

With the widespread uses of GPS equipments, the residents rely more and more on various map navigation applications for their daily travels. It poses the requirement for map updates to ensure the high quality of the map. As one of the main reasons caused the change of road network, road closures come up very often due to the events like temporary traffic control, marathon, and road under repair, etc. Delayed or missed detection of road closure not only creates inconvenience for travel, but results in relatively great economic losses of road users. For instance, on October 10, 2019, the bridge collapsed on National Highway 312, Wuxi City [1] caused the road under the bridge to be closed and the traffic jams to be occurred in its surrounding roads for a long time.

A tremendous amount of trajectories are collected by different kinds of positioning devices, which provides us an opportunity to update the map. In recent years, the existing map update researches focus more on the detection of the missing roads, less on the detection of road closure events [1]. Due to the challenges stemming from dynamic changing of the road network as well as the interplay in between the adjacent roads' traffic situations, and skewed distribution nature of data, only a few research have been conducted on the issue of road closure detection [17, 21, 22, 31]. The aforementioned methods identify the road closure in terms of the plunge in traffic flow, but which may lead to error detection of traffic jams having the same property. In addition, they are unfit for detecting one side closure of two-way road and the closure that

[1] http://www.chinanews.com/sh/2019/10-10/8975525.shtml

occurred on the middle part of the road. As illustrated in Figure 1, both are the cases of road closures due to road construction. In Figure 1(a), the closure in the middle of the road makes the vehicles have to take a U-turn towards the original road, which leads to an increase in the frequency of U-turns on the road. In Figure 1(b), one side of the road is completely closed to traffic but the other direction lane can still be passed normally, accompany with which is the reduction of the amount of the turning volume of adjacent roads. But in both cases, the traffic flows of the closed roads do not completely drop to zero.

In the light of the above observations, we present a two-phase road closure detection framework based upon multi-feature fusion, called *RCDM*. At the off-line road closure feature modeling phase, we partition the road network into grids. Based upon the analysis of historical data, we first extract the correlation strength of traffic situations among all roads. Then we obtain three important features that will be used for online detection, which includes traffic flow of all the grids, turning volume from adjacent roads and U-turn frequency in each time period. At the online phase, we design a two-step detection mechanism consisting of closed grid candidate detection and road closure identification. In view of the spatio-temporal dependency of the road closure feature, deep learning technique is introduced into predicting the values for closure features of each grid (or road) in the current time period [25, 29]. Through calculating the gaps between the predictions and true values for closure features of each grid (or road), and comparing them with the predefined thresholds, the closed grid candidates (or road closures) are found.

To summarize, this paper makes the following contributions: (1) To the best of our knowledge, ours is the first work to detect road closure based upon multiple features, which is composed of off-line feature modeling and online identification of road closures. (2) To improve the effectiveness of online detection, deep learning technique (including the combination of *CNN* and *LSTM* and that of *GCN* and *LSTM*) is leveraged to predict the values for multiple road closure features. (3) Experimental results on real taxi trajectory data from Chengdu, Shanghai and Beijing show our proposal performs better than the state-of-the-art methods.

The remainder of this paper is organized as follows. Section 2 reviews the most related work. The preliminary concepts are introduced in Section 3 and the detail of *RCDM* framework is outlined in Section 4. Section 5 presents the experimental results, followed by conclusions in Section 6.

2 RELATED WORK

Traffic anomaly detection. Recently, as one of the vital components of map update, road closure detection has attracted wide widespread attentions from academia and industry. Some researchers also regarded road closure as a type of traffic anomaly, and detected it according to abnormal changes in traffic situations within the regions [2, 8, 11, 12, 16, 26, 30]. Chawla et al. employed Principal Component Analysis (or *PCA* for short) to mine traffic anomalies based on the deviations of traffic flow from historical traffic profiles [2]. Kuang et al. combined wavelet transform with *PCA* to detect traffic anomalies in urban regions due to low (or high) rates of change in traffic flow [8]. Liu et al. proposed an anomaly detection model by building a region graph firstly and then detecting the traffic flow outliers by calculation of *Mahalanobis Distance* [11]. Pang and Wang et al. used Likelihood Ratio Test (or *LRT* for short) statistic method to identify the contiguous set of regions and time intervals that have the largest statistically significant deviation from expected behavior based on the number of taxis in the regions [16, 30]. Zhang et al. proposed a two-step method to detect urban anomalies by employing Support Vector Machines method [26]. Luo et al. adopted the relative divergence to calculate abnormal fluctuation of each region [12]. Zhang et al. proposed an isolation-based abnormal trajectory detection method. It can screen out the traffic flow with sudden change by comparing abnormal routes with historically normal routes and identify it as road closure [3].

Road closure detection. Due to dynamic changing of the road network and skewed distribution of trajectory data, and interactions in between the adjacent roads' traffic situations, only a few research have addressed the issue of road closure detection. They can be grouped into two categories: threshold-based and statistics-based methods [21, 22, 31]. Wang et al. put forward a threshold-based method to learn the upper and lower thresholds from historical data [22]. It identified the road closure by detecting the abnormality of traffic flow sequence in the time dimension. Stanojevic et al. introduced the betweenness centrality to indicate the importance of roads in the cold start phase and detected the road closure among those important roads [21]. Zhu et al. aggregated roads with a node degree less than 3 into a road set and determined the road closure based on the variation in the ratio of the road flow rate to the road set flow rate [31]. Because of the limitation of a fixed threshold, the accuracy of the threshold-based method is low especially when the trajectory data is skewed in the road network. Pietrobon et al. proposed a statistical-based method to detect road closure by establishing a Poisson process [17]. But it has lower detection precision due to that it ignores spatio-temporal correlation in between roads.

Traffic prediction. R. Fu et al. leveraged RNN and LSTM to predict traffic flow and travel time, and achieved better prediction performance due to the use of the self-circulation mechanism in learning temporal dependence [6, 18]. Since the change of traffic is constrained by the urban regions and roads, it is important to capture the spatial and temporal dependences for traffic prediction. To take full advantage of spatial features, Wu and Tan proposed an architecture called CLTFP. It predicts traffic flow by combining a 1-dimensional CNN with two LSTMs, which mine spatial dependence and short-term variability respectively [24]. However, CNN is limited in capturing features (e.g., images and regular grids) of Euclidean space, which enables CNN cannot be used for complex road network structures. Recently, some studies converted grid-based CNN to graph-based structure for traffic prediction. Seo et al. introduced graph convolutional recurrent network into prediction [20]. Zhao and Li combined gated recurrent units with graph convolution to predict traffic respectively [10, 29]. Zhang et al. presented a prediction method based on graph attention [27].

The aforementioned methods of traffic anomaly detection and traffic prediction had one problem in common: they detected road closure according to the variation of traffic flow. But in actual applications, they may misjudge congestions as road closures. More importantly, they cannot be used to identify some special types of road closures such as one-way closure, closure caused by lanes

merging and middle closure within the road. As a result, it necessitates a high-efficiency road closure method based on multiple features to identify all road closures promptly and accurately. Furthermore, to proliferate the detection precision, it needs to consider spatio-temporal dependence and design appropriate spatial capture models fitting for different prediction objects (e.g., grids and roads) when predicting the values of road closed features.

3 ROAD CLOSURE DETECTION

In this section, we introduce some preliminary concepts and formally define the issue of road closure detection based on trajectory data.

Definition 1 (*Road Map*) : A road map is typically represented as a graph $\mathcal{M} = (\mathcal{V}, \mathcal{E})$, where the set of vertexes (denoted as $\mathcal{V}$ and $v \in \mathcal{V}$) includes all roads. The set of edges (denoted as $\mathcal{E}$) represents the reachability relationship among the roads.

Definition 2 (*Trajectory*) : A trajectory $\mathcal{T}r$ refers to a series of position points generated by a moving object in chronological order, denoted as $\mathcal{T}r =< tp_1, tp_2, ..., tp_n >$. Each position point $tp_i =< lng_i, lat_i, t_i, id_i >$, where lng_i and lat_i denote the longitude and latitude of one positional point at timestamp t_i respectively, id_i denotes which vehicle this trajectory belongs to.

To find the road closures efficiently, we design the online detection mechanism consisting of closed grid candidate identification and closed roads detection.

We first partition a city into grids according to the longitude and latitude, and get a grid map (denoted as $\mathcal{G}$). Then we obtain the matched road sequences by map matching on trajectory data.

Definition 3 (*Closed Grid Candidate*) : Given a set of grids and a traffic flow outlier threshold th_f, a grid is identified as a closed grid candidate $\mathcal{GC}$ iff the gap between the predicted value and the true value of the gird's traffic flow beyonds th_f.

In doing so, a detection area, denoted as $\mathcal{DA}$, is composed of $\mathcal{GC}$ and surrounding grids.

Definition 4 (*Correlation Strength*) : Given a set of matched road sequences, the correlation strength between any two adjacent roads, denoted as $\mathcal{CS}$, is the frequency of such two roads co-occurring in all matched road sequences.

For any target road, its adjacent road with maximal correlation strength is regarded as its strongly correlated road.

Definition 5 (*Closed Road*) : Given a turning volume outlier threshold th_t and a U-turn count outlier threshold th_u, a road is identified as a closed road $\mathcal{VC}$, iff the gap between the prediction and the true value of the turning volume of $\mathcal{VC}$'s strongly correlated road (or U-turn count of $\mathcal{VC}$) beyonds th_t (or th_u).

Finally, we summarize the problem as below.

Problem Statement : Given a set of roads and a collection of trajectories, the goal is to find the *Closed Road*($\mathcal{VC}$) based on sharp drop of traffic flow as well as the drop of turning volume of $\mathcal{VC}$'s strongly correlated roads and the increment of U-turn count of $\mathcal{VC}$.

4 OVERVIEW OF *RCDM* FRAMEWORK

We propose a Road Closure Detection framework based upon Multi-feature Fusion (or *RCDM* for short) using trajectory data. As shown in Figure 2, *RCDM* is composed of *Off-line Road Closure Feature Modeling* phase and *Online Road Closure Identification* phase. In the off-line phase, we first partition the region into grids, and then extract road correlation strength and road closure features from historical data. In the online phase, we screen out closed grid candidates based on a significant drop of traffic flow within grids. And on this basis we try to find closed roads according to the drop of strongly correlated road turning volume or the increment of U-turn frequency.

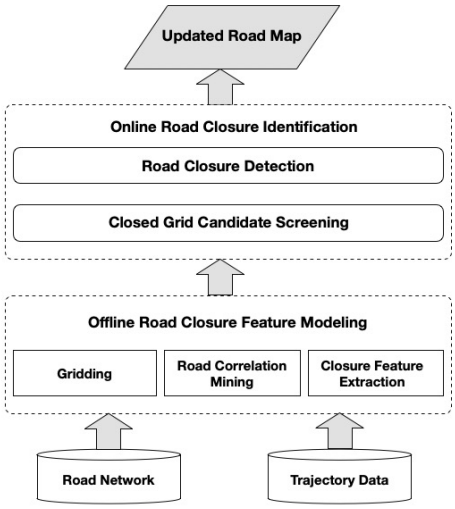


Figure 2: The Framework of *RCDM*

4.1 Off-line Road Closure Feature Modeling

It is observed that the road closure is not an isolated event. Specifically, the closure of any road would affect the traffic situations of its neighboring regions. Given that, in the off-line modeling phase, we first split the region to be detected into grids and attempt to extract the correlation strength of the roads. Based upon historical data, we try to extract a series of road closure features for online detection purposes. They include traffic flow of grids as well as turning volume of strongly correlated roads and U-turn frequency of targeting road, as is depicted in Algorithm 1.

Initially, we split the region to be detected into a set of grids G of equal size (at line 2), as shown in Figure 3. Then we number each grid according to its row and column, and on the basis of which we build the inverted index for the road network and trajectory data respectively. Adopting the grid division method has the added benefit of reducing online detection overhead when screening out closed grid candidates. In addition, consider that the average length of roads and excessively large grid may ignore the fluctuation of local trajectories, we set the grid's side size to 500 meters after testing on real data.

Correlation Strength Mining for the Roads. Each road's traffic situation usually would be affected by its upstream roads (also called correlated roads). We attempt to count the frequency of the vehicle successively travels across any two adjacent roads to extract their correlation strength (denoted as $\mathcal{CS}$) based on historical trajectory data (at lines 3-4). To be specific, we first employ

Algorithm 1: Off-line Road Closure Feature Modeling

Input: The set of trajectories $\mathcal{T}r_s$, the road map $\mathcal{M}$, the range of detection region $\mathcal{DR}$;
Output: the hour-level traffic flow time series of each grid $\mathcal{TF}_h$, the hour-level U-turn frequency and traffic turning volume time series of each road $\mathcal{RU}_h$,$\mathcal{TV}_h$;

1 $\mathcal{S} \leftarrow \emptyset$, $\mathcal{G} \leftarrow \emptyset$; //$\mathcal{S}$ denotes the matched road sequences by map matching
2 $\mathcal{G} \leftarrow gridding(\mathcal{DR})$; //split the regions to be detected into grids
3 $\mathcal{S} \leftarrow mapMatching(\mathcal{T}r_s, \mathcal{M})$;
4 $\mathcal{CS} \leftarrow getCorrelationStrength(\mathcal{S})$;//obtain the correlation strength of the roads in the aspect of traffic situation
5 **for** g **in** $\mathcal{G}$ **do**
6 $\quad \mathcal{TF}_h \leftarrow getHourTrafficFlow(g, \mathcal{T}r_s)$;
7 **end**
8 **for** *each hour* **do**
9 $\quad \mathcal{RU}_h \leftarrow \mathcal{RU}_h \cup UturnFrequencyExtraction(\mathcal{T}r_s, \mathcal{M})$
10 **end**
11 **for** *each hour* **do**
12 $\quad \mathcal{TV}_h \leftarrow \mathcal{TV}_h \cup getTrafficTurning(\mathcal{S}, \mathcal{CS})$
13 **end**
14 **return** $\mathcal{TF}_h, \mathcal{RU}_h, \mathcal{TV}_h$

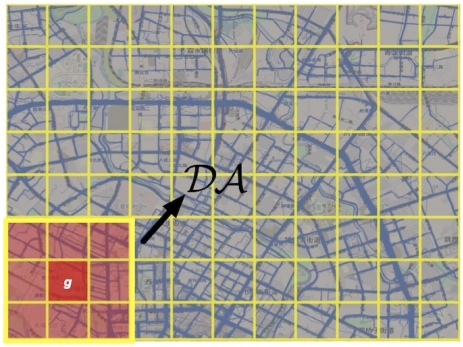


Figure 3: Gridding

Hidden Markov-based map matching algorithm [15] to obtain the matched road sequences (denoted as $\mathcal{S}$) for each trajectory. Subsequently, we count the number of any two successive roads co-occurring for all trajectories to derive the correlation strength of such a road pair. For instance, when a trajectory $\mathcal{T}r$ is matched to the road sequence $\mathcal{S} =< v_1, v_2, v_3, v_4 >$ by map matching, the number of occurrences of any road pair $< v_1, v_2 >$ which the vehicle successively travel across would be counted. By this way, the correlation strength between road v_1 and its adjacent road can be obtained through analyzing all trajectories. Further, the adjacent road with maximal correlation strength is regarded as v_1's strongly correlated road. For any two-way road, there is a strongly correlated road in each direction. Comparing with other related roads, the changes in turning volume of strongly correlated road after closure are more dramatic. Hence, strongly correlated road is enough to



Figure 4: Illustration of Roads' Correlation Strength

capture traffic changes during road closure. As illustrated in Figure 4, for the target road highlighted in yellow line, its adjacent roads have different correlation strengths, i.e. the correlation strength gradually increases from gray line to red one.

Road Closure Feature Extraction. To extract closure features for each grid (or road) from historical trajectory data, we first partition the trajectory $\mathcal{T}r$ into segments at the hourly level. Then we count the number of vehicles that travel across each grid to obtain its normal hourly traffic flow statistics (denoted as $\mathcal{TF}_h$) (at lines 5-7).

We propose an algorithm called $UturnFrequencyExtraction$ (at lines 8-10) to obtain normal frequency of U-turn for all roads in each hour based on historical data. To be specific, we first calculate the angle gap between each position point and its previous one for each trajectory to identify the turning points. If that angle gap is greater than preset angle threshold th_{angle1} (here th_{angle1} is set as 35), such a position point is marked as a turning point. We leverage *DBSCAN* [4] method to cluster all turning points and represent the clustering results in the form of a convex hull. *DBSCAN* doesn't need to predefine the number of classes. It is insensitive to noise and can find clusters of different shapes. Then we treat the first point of a certain trajectory entering the convex hull as entry point and the last point leaving the convex hull as leaving point. If the angle gap between entry point and leaving point is greater than the preset angle threshold th_{angle2} (here th_{angle2} is set as 150), the trajectory is judged as a U-turn trajectory. By map matching, we obtain the roads where the U-turn trajectories locate, and get normal frequency (denoted as $\mathcal{RU}_h$) of U-turn occurring on those roads. As illustrated in Figure 5, a convex hull is marked by a green irregular oval, in which contains some turning points (highlighted in yellow filled dots). A trajectory (highlighted in red thick line) whose entry point and leaving point has a relatively large angle gap is viewed as a U-turn trajectory.

Moreover, we try to obtain normal turning volume of strongly correlated road every hour. We first obtain the matched road sequence (denoted as $\mathcal{S}$) for each trajectory through map matching. Then we screen out the strongly correlated roads for any road by correlation strength mining. By counting the number of occurrences of the road pair which the vehicle successively travels across, we can get the turning volume (denoted as $\mathcal{TV}_h$) for any strongly correlated road of each road in each hour (at lines 11-13).

4.2 Online Road Closure Identification

Although the traffic situation has periodicity and trending (or known as temporal dependence), e.g., morning rush hour and

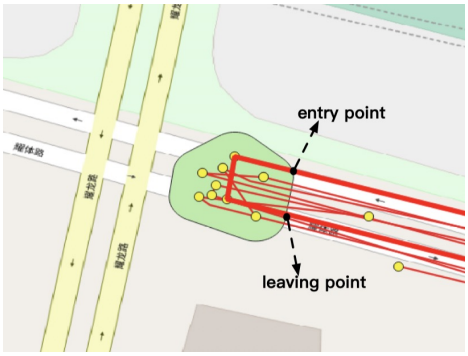


Figure 5: U-turn Extraction

evening peak hour, the road network is dynamically changing. It is likely to misjudge the event like a traffic jam as road closure by comparing the values of closure features with historical normal values to pinpoint the outliers. As stated previously, any road closure will have an impact on the traffic situation of its surrounding area (otherwise known as spatial dependence property). Given this, at the phase of online detection, we first leverage deep learning technique to predict the values of closure features for each gird or road in the current time period instead of using their historical normal values. Based on that, we compare true values with the predictions to find the grids or roads that have significant differences on closure features in the current time period, as depicted in Algorithm 2. In our prediction model, complex spatial-temporal dependence of trajectory data is taken into account. To achieve the goal of identifying closed grid candidates first and then the road closures, we adopt the strategy consisting of two different combinations of neural network models to extract spatial-temporal dependence from two region levels (i.e. grid and road) respectively, as shown in Figure 6.

Closed Grid Candidate Screening. In an actual road network, the traffic flow of each road section is usually fixed in every time period, which is essentially determined by the length of the road and the speeds of the passing vehicles. Consider the effects of the road closure on its neighboring roads' traffic situations, we first screen out the grids which may contain road closure according to dramatically drop of their traffic flows. Doing this has another advantage of reducing the costs, due to that it does not implement map matching at the beginning of online detection.

We adopt the combination of *CNN* and *LSTM* [25] models to predict the values of closure features in the current time period. If the gap between the prediction and the true value exceeds the predefined outlier threshold, the grid will be judged as a closed grid candidate. *CNN* is a biologically inspired model that has been widely used to process images, speech and time series [28]. It is well known that the grid is a regular spatial structure (i.e. Euclidean structure) having translation invariance, and the change of traffic flow in each grid is mainly affected by surrounding grids. A Convolutional Neural Network (*CNN*) with a fixed-size convolution kernel is fit for efficiently capturing spatial correlation of the data that can be expressed as a two-dimensional matrix. To be specific,

Algorithm 2: Online Road Closure Identification

Input: The set of trajectories in real time $\mathcal{T}r_s$;
Output: a set of closed roads $\mathcal{VC}$;
a set of closed grid candidates: $\mathcal{GC} \leftarrow \emptyset$;
a set of closed roads: $\mathcal{VC} \leftarrow \emptyset$;
for g **in** $\mathcal{G}$ **do**
  $\mathcal{TF}_c \leftarrow getCurrentTF(g)$
  $\mathcal{E}_f \leftarrow getEastimate(\mathcal{TF}_h)$ // *CNN* and *LSTM* model
  if $(\mathcal{E}_f - \mathcal{TF}_c) > th_f$ **then**
    $\mathcal{GC} \leftarrow \mathcal{GC} \cup g$
  end
end
$\mathcal{DA} = getDetectionAreas(\mathcal{GC})$
for da **in** $\mathcal{DA}$ **do**
  for v **in** $\mathcal{M}(da)$ **do**
    $\mathcal{TV}_c \leftarrow getCurrentTV(\mathcal{T}r_s(da), \mathcal{M}(da), \mathcal{CS}))$ // map matching
    $\mathcal{RU}_c \leftarrow getCurrentRU(\mathcal{T}r_s(da), \mathcal{M}(da)))$// *DBSCAN*
    $\mathcal{E}_t \leftarrow getEastimate(\mathcal{TV}_h)$ // *GCN* and *LSTM* model
    $\mathcal{E}_u \leftarrow getEastimate(\mathcal{RU}_h)$ // *GCN* and *LSTM* model
    if $(\mathcal{E}_t - \mathcal{TV}_c > th_t)$ **or** $(\mathcal{RU}_c - \mathcal{E}_u > th_u)$ **then**
      $\mathcal{VC} \leftarrow \mathcal{VC} \cup v$
    end
  end
end
return $\mathcal{VC}$

historical hourly traffic flow of each grid is turning into an image-like [28] matrix X, which is a two-dimensional matrix containing traffic flow. In view of that a convolution only considers close spatial correlation property, we design a *CNN* with multiple layers. The conversion of each layer is defined as:

$$X_{i,t}^{k} = f(W^{k} * X_{i,t}^{k-1} + b^{k}) \tag{1}$$

where $*$ denotes convolutional operation, and f is ReLU activation function. ReLU function is preferred over traditional sigmoid/tanh functions because: a) it is more computationally efficient, b) it can help avoid gradient explosion and vanishing problems [14]. $X_{i,t}^{k}$ represents the output of the i area of the k layer at time period t, and W, b are learned parameters. After K convolution layers, we use a flatten layer to transform the output as a $1-D$ feature vector with spatial attributes and send it through a fully connected layer to output the final result.

Since the present traffic flow is affected by the traffic condition of the previous moment or even earlier, it is necessary to capture the temporal dependence of traffic flow. To this end, we employ *LSTM* [5] model, which uses a gated mechanism to memorize as much long-term information as possible. It is widely applied to address the exploding and vanishing gradient issue of traditional Recurrent Neural Network [19]. With this model, the main component of the Long short-term memory layer is the memory cell which records

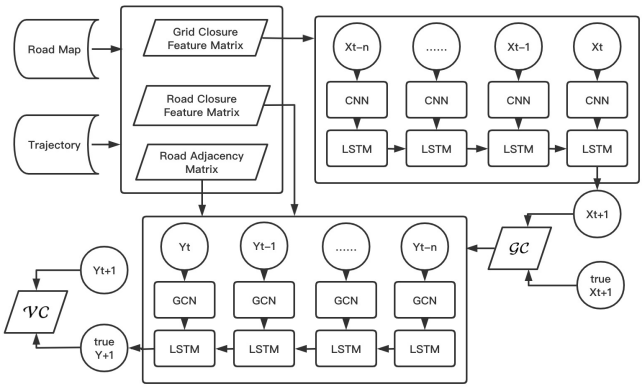


Figure 6: Prediction Model

the state of every cell. A memory cell consists of four elements: the updated value of cell's state, an input gate, a forget gate and an output gate. They determine how much previous and current information needs to be kept and forgotten.

In addition, we take into account the periodic changes of the day and week level. *LSTM* covers the feature data of the previous seven days.

The final output contains both spatial and short-term temporal information, which is obtained by a fully connecting layer. Based on the extracted historical hourly traffic flow $\mathcal{TF}_h$, we predict the traffic flow $\mathcal{E}_f$ of each grid in current time period. If the difference between the prediction and the true value of traffic flow of some grid in current time period is greater than the traffic flow outlier threshold th_f, that grid is marked as a closed grid candidate which may have road closure. By doing this, we obtain a set of closed grid candidates $\mathcal{GC}$ (at lines 3-9). In view of skewed distribution of trajectory data in road network, the value of th_f should be adaptive. We adjust its value in terms of the trajectory distribution density d_g within the target grid g by using the method proposed in [13]. It can be expressed as:

$$th_f = \frac{d_g}{avg(d_g)} \times \alpha \tag{2}$$

here $avg(d_g)$ is the average density of trajectory data within each grid in historical data, and α is a learned threshold evaluated by the experimental part.

Consider that some road closure events occur where two or more girds meet, we extract every candidate closed grids and its adjacent grids to form a road closure detection region (denoted as $\mathcal{DA}$, at line 10). As shown in Figure 3, a candidate closed grid g and its neighboring grids are formed into a region highlighted in a red square.

Road Closure Detection. To obtain the values of road closure features in current time period, the road network data $\mathcal{M}(da)$ and the trajectory data $\mathcal{T}r_s(da)$ in each candidate detection region are obtained respectively based on inverted index structures. Then we get the matched roads sequence for each trajectory by map matching [15]. According to the correlation strength of each road $\mathcal{CS}$, we get the turning volume of the strongly correlated road of each road in current time period $\mathcal{TV}_c$. In addition, we get the frequency of U-turn $\mathcal{RU}_c$ in current time period based on clustering for turning points (at lines 13-14).

Actually, the traffic situation of each road is not only affected by its neighboring roads, but also influenced by that within previous time periods. In other words, there are spatial-temporal dependencies on traffic situations of the roads. Given that, we still adopt the combination of neural network models to capture spatial dependency and temporal dependency respectively. As road network is in the form of a graph (non-Euclidean structure), the ordinary *CNN* model cannot obtain its complex spatial topological characteristics well due to its fixed-size convolution kernel. As a result, Graph Convolutional Neural Network (*GCN*) takes the place of *CNN*, and works with *LSTM* [29] to capture spatial-temporal dependency of the roads' traffic situations. To be specific, we first establish adjacency matrix A for the roads based on their upstream and downstream relationships with the adjacent roads. Each node contains two attributes, which includes turning volume of strongly correlated road and U-turn frequency. Then we construct the feature matrix for these two features respectively, denoted as Y. *GCN* model bridges spectral graph theory and deep neural networks by constructing a filter in *Fourier* domain. The filter acts on the nodes of the graph and captures spatial features between the nodes through their first-order neighborhoods. *GCN* with multiple convolutional layers is defined as below:

$$H^{l+1} = \sigma(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^l W^l) \tag{3}$$

where $\tilde{A} = A + I$ is the matrix with added self-connections and I is the identity matrix. $\tilde{D}$ is the degree matrix of $\tilde{A}$, H^l is the output of l layer, W^l is a trained weight matrix and σ is a nonlinear activation function. Then we construct a two-layer *GCN* [7] whose overall forward propagation formula is defined as:

$$Z = f(Y, A) = softmax(\hat{A}ReLU(\hat{A}YW^{(0)})W^{(1)}) \tag{4}$$

where $\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$ acts as pre-processing, the activation function uses *ReLU* and *softmax* respectively. $W^{(0)}$ is the weight matrix from input to hidden layer and its size is the length of feature matrix times the number of hidden unit. $W^{(1)}$ is the weight matrix from hidden layer to the output layer, whose size is the number of the hidden unit times the prediction length. We use *GCN* model to obtain topological relationship between the target road and its adjacent roads, and encode the topological structure as well as road attributes of the road network. Subsequently, the result with spatial property is also input into *LSTM* for capturing temporal attributes.

After we obtain the prediction $\mathcal{E}_t$ based on $\mathcal{TV}_h$, we calculate the gap between prediction $\mathcal{E}_t$ and the current $\mathcal{TV}_c$. When that gap is greater than the turning volume outlier threshold th_t, a closed road $\mathcal{VC}$ is identified. Further, a road can be verified as one-way or two-way based upon *OSM*[2] metadata. As for the case that one side of a two-way road is closed, it is recognized when the gap between prediction $\mathcal{E}_t$ and the current $\mathcal{TV}_c$ with one direction of the road is greater than th_t but another is not (at lines 15,17). It is observed that U-turn frequency will increase when the middle part of the road is closed. We obtain the prediction $\mathcal{E}_u$ based on $\mathcal{RU}_h$, and judge the road as closure when the gap between the prediction

[2]http://wiki.openstreetmap.org/

$\mathcal{E}_u$ and true value $\mathcal{RU}_c$ is larger than the U-turn outlier threshold th_u (at lines 16,17). Here th_t and th_u are adaptive thresholds and they can be expressed as:

$$th_t = \frac{avg(\mathcal{TV}(v))}{avg(\mathcal{TV})} \times \beta \tag{5}$$

where $avg(\mathcal{TV})$ is the hourly average $\mathcal{TV}$ of all roads in historical data and $avg(\mathcal{TV}(v))$ is the average of the road v.

$$th_u = \frac{avg(\mathcal{RU}(v))}{avg(\mathcal{RU})} \times \gamma \tag{6}$$

where $avg(\mathcal{RU})$ is the hourly average $\mathcal{RU}$ of all roads in historical data and $avg(\mathcal{RU}(v))$ is the average of the road v. β and γ are the learned thresholds evaluated in subsequent experiments.

5 EXPERIMENTAL EVALUATION

In this section, we conduct extensive experiments to evaluate the effectiveness and efficiency of *RCDM*. We first describe three real trajectory datasets used in this paper. Then we present the evaluation metrics for the experimental results. After that, we provide a comparison of the effectiveness and efficiency between different baseline methods and *RCDM*. Finally, a set of real case studies and the ablation experiment are presented to demonstrate the effectiveness of our proposal. All experiments are conducted on a 5-node clustering running Spark 2.2.0 over Ubuntu 12.0.4. Each node is equipped with 8 cores Intel E5335 2.0 GHz processor and 16 GB memory. The program and baselines are implemented in Python 3.7.

5.1 Data Description

We choose the data sets from three major cities for evaluating our experiments, which can represent complex road network of most cities in China. It contains the taxi trajectory data set of 2018 in *Shanghai* [3] (or *Shanghai*2018 for short), the taxi trajectory data set of 2013 in *Beijing* (or *Beijing*2013 for short) and the taxi trajectory data set of 2018 in *Chengdu* (or *Chengdu*2018 for short). The purpose of this is to cover more traffic scenarios in different places, and the parameters of detection models can be more applicable to other different cities or countries. In addition, we choose an open source map *OpenStreetMap (OSM)* as our map data.

*Shanghai*2018. It is a public data set derived from an innovation and application competition (called *Shanghai Open Data Apps*, or *SODA* for short). *SODA* is an annual event that enables data analysts to utilize a massive amount of data to develop useful *apps* to tackle some city issues. *Shanghai*2018 is the taxi trajectory data set of *Shanghai Qiangsheng* Taxi company from April 1 to April 30, 2018. There are about 200,000 trajectories and 100 million trajectory points every day. The average sampling frequency is 10 seconds. Each trajectory point contains the attributes such as vehicle number, timestamp, longitude and latitude, etc.

*Chengdu*2018. It contains the trajectory data of *Chengdu* obtained from *Didi* platform in October and November 2018. *Didi* generates more than 106TB of vehicle trajectory data on a global scale and handles more than 40 billion route requests every day. *Chengdu*2018 contains about 2 billion trajectory point coordinates, more than 10 million trajectories, about 24.7G, and the average sampling frequency of which is 3 seconds. Each trajectory point contains four attributes: driver ID, timestamp, longitude and latitude, etc.

*Beijing*2013. It is a taxi trajectory data set from October 1, 2013 to November 1, 2013 in *Beijing*. There are about 50,000 trajectories every day, and the average sampling frequency is 10 seconds.

OSM data. Road network of *Beijing* is from *OSM* with 150,183 nodes and 169,294 edges, road network of *Shanghai* is from *OSM* with 191,684 nodes and 210,909 edges, and road network of *Chengdu* is from *OSM* with 168,216 nodes and 187,444 edges.

As the sampling intervals of three data sets are different, to ensure conditional consistency, we first subsample the *Chengdu*2018 data set to keep the sampling interval consistent. We also use 60% of the data (not having road closures) from three data sets as training data for building the prediction models. In addition, in order to ensure the fairness and accuracy of efficiency evaluation compared with the baseline methods, we take the trajectories equably from three datasets for the efficiency experiment.

5.2 Evaluation Method

We use three evaluation metrics (i.e. *Precision*, *Recall* and *F1-score*) to evaluate our effectiveness and ablation experiments, and evaluate the efficiency experiment by the runtime. Let TP denote the number of the closed roads correctly identified by the detection algorithm, FP denote the number of the closed roads incorrectly identified by the detection algorithm, and FN denote the number of the closed roads in the manually labeled road closure set but unidentified by the detection algorithm. *Precision*, *Recall* and *F1-score* are calculated respectively as follows:

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{9}$$

Precision indicates how many of the detected road closures that coincide with the ground truth closures, *Recall* indicates how many of the ground truth closures that have been successfully identified by the detection algorithms, and *F1-score* is the weighted harmonic average of *Precision* and *Recall*.

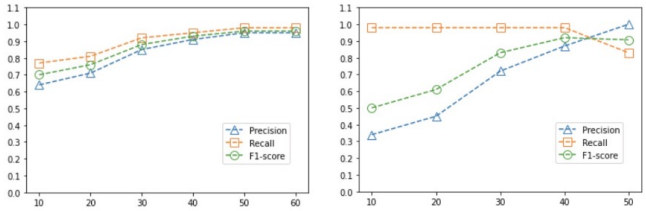
5.3 Effectiveness Evaluation

In this subsection, we evaluate the effectiveness of *RCDM*. We first describe the process of manually labeling data, then introduce the process of threshold selecting for the detection phase. Finally, we compare our proposal with three baseline methods.

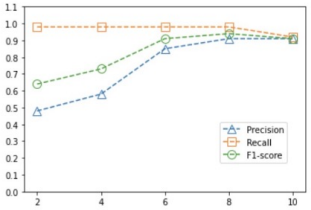
Lable Extraction. According to the temporal and spatial scope of three data sets, we collect 65 road closure events through news, government announcements and traffic management websites. They include but are not limited to the closed roads due to major sporting and entertainment events, traffic accidents, road repair events and political events, etc.

[3] http://shanghai.sodachallenges.com/

Threshold Selection. In our implementation, we have tried possible thresholds of grid's traffic flow, strongly correlated road turning volume and U-turn frequency. To determine most appropriate thresholds, we verify the detection result with the ground truth road closures. *Precision*, *Recall* and *F1-score* are calculated separately to reflect the effectiveness of our proposed method when setting different thresholds.



(a) Varying α (b) Varying β



(c) Varying γ

Figure 7: Parameters Evaluation

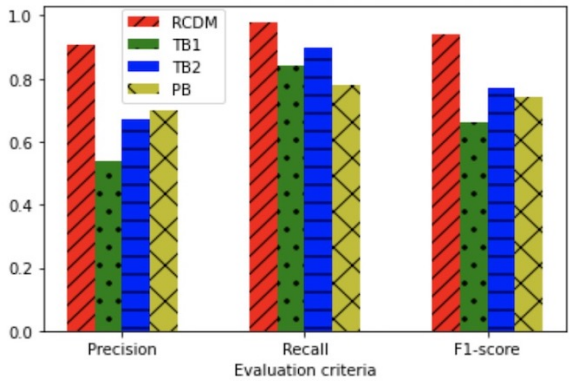
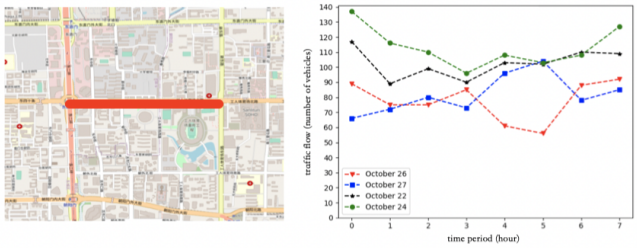


Figure 8: Effectiveness Evaluation

As illustrated in Figure 7, when α, β and γ continue to increase, *F1-score* reaches a bottleneck at 50, 40 and 8 respectively. Therefore, we determine that *RCDM* achieves the best performance when α = 50, β = 40, γ = 8 after repeatedly tuning. It should be noted that larger β and γ will miss some road closures and hence leads to the decline of *Recall*, so we cannot set them too large.

Comparing with Baselines. In this experiment, we compare *RCDM* with three approaches to verify the superiority of *RCDM*:



(a) Congestion of Beijing Workers' Stadium (b) Trend of traffic flow

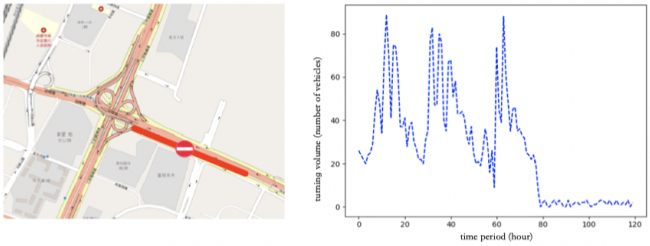
Figure 9: Case of Congestion

(i) *threshold*-based method [31] (or *TB*1 for short), which aggregates roads with a node degree less than 3 into a road set, and determines road closures based on the change in the ratio of the road flow rate to the road set flow rate.

(ii) Cold-start road closure detection method [21] (or *TB*2 for short), which introduces the definition of the *betweenness centrality* to indicate the importance of roads in the cold start phase and determines the closure based on the time interval of the vehicle passing through the road.

(iii) *Poisson process*-based method [17] (or *PB* for short), which is a statistical-based method to determine road closure by establishing a *Poisson process* and run a *p-value test* on a *chi-square distribution*.

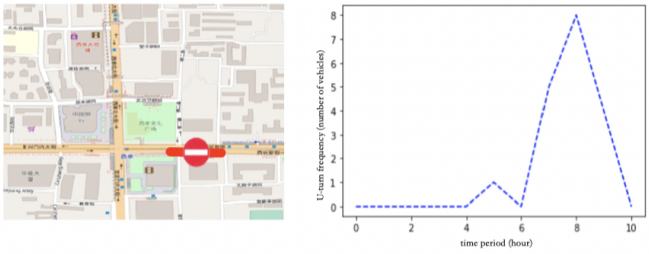
As illustrated in Figure 8, *RCDM* has a good performance except for the regions where trajectory distribution is sparse. Next, we illustrate the effectiveness of our proposal with many real-life examples.



(a) Closure in Baohua Road (b) Trend of turning volume

Figure 10: One Side Closure

It can be seen that the baseline methods detect road closure only according to dramatically drop of traffic flow and they simply record the time interval of vehicles passing through the road. As shown in Figure 9, a concert in Beijing Workers' Stadium was held on October 26, 2013 at 18:30 pm, which caused a rapid decline in traffic flow due to extreme congestion. Therefore, they are easy to misjudge congested events and cannot identify some special road closure events. In addition, our proposal can detect one side closure of the two-way road and the closure of the middle part of the road. As shown in Figure 10, due to the construction of Chengdu Baohua Road on October 31, 2018, the turning volume of single direction



(a) Closure in West Changan Avenue

(b) Trend of U-trun frequency

Figure 11: Closure in the Middle of the Road

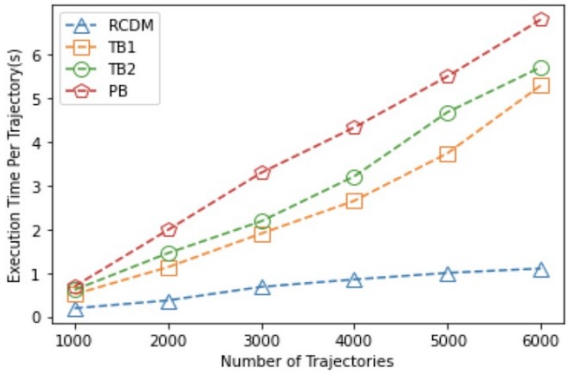


Figure 12: Efficiency Evaluation

on its strongly correlated road suddenly drops to 0. Also, as shown in Figure 11, there is a marathon held near Beijing West Changan Avenue on October 20, 2013. It can be seen that U-turn frequency on this road at first has a sudden increment, and then returns to its original state after the road closure. The above road closures cannot be identified only through the sharp drop detection of traffic flow.

5.4 Efficiency Evaluation

Since the amount of trajectory data is different according to the change of area or time during online detection, we choose different number of trajectories to evaluate the efficiency of our proposal. As shown in Figure 12, the running time of our proposal increases as the number of trajectories increases. When the number of trajectories is small, the gap between *RCDM* and baseline methods is not obvious. But when the number of trajectories continues to increase, the execution time of *RCDM* is much lower than the comparison algorithms. This is because *RCDM* first screens out the closed grid candidates, which reduces the computational cost on the detection of specific closed roads. In summary, *RCDM* can run efficiently in dealing with a large amount of trajectory data, which shows that our method has better performance.

5.5 Case Studies

We choose two special closure cases in the experiments: periodic closure and road closure caused by merging lanes. As illustrated in Figure 13(a) and (b), there is a flag-raising event occur at *Tiananmen Square* in East Changan Avenue. It regularly caused significant drop in traffic flow at about 6:00 in the morning because of the road closure. In addition, some freight trains will pass through a few urban roads at a fixed time period every day, which makes these roads closed for a short period of time, as shown in Figure 13(c).

The closure caused by merging lanes is a particular situation of the one side closure, as illustrated in Figure 14(a). For some cases of one side closure, we observe that turning volume of strongly correlated road has not declined largely, but their trajectories have obvious deviation compared to historical normal route. As a result, we analyze these examples separately in order to detect this type of road closure.

The preliminary data analysis and daily observation show that this type of closed road has a reduction in traffic flow and turning volume of adjacent roads, except no significant change for U-turn frequency. We attempt to calculate the offset of the trajectories on the road to detect the closure caused by merging lanes. This type of road closure shows the phenomenon is that the original two lanes are changed to a single lane, which leads to the trajectories lean toward the opposite side of the road. To address this, we need to extract the average offset of each road's matched trajectories from historical data. Then we inference the centerline of each road and obtain the set of trajectories on the road by map matching. The points in each trajectory are traversed and its vertical distance from the road centerline is regarded as the offset of each point. Finally, we extract the average offset of each road's matched trajectories O_h.

In detection phase, we first calculate the offset of current trajectory on the road O_c based on the road centerline. Then we calculate the *Wasserstein* distance [23] by calculating the average offset distribution of historical trajectory O_h and the offset distribution of the trajectory to be detected in current time period O_c. *Wasserstein* distance is also called Earth Mover's distance, which is a measure of the distance between two distributions. For example, the offset of $\mathcal{T}r_1$ is $O_i^{\mathcal{T}r_1} = \{p_1, p_2, ..., p_n\}$ and the offset of $\mathcal{T}r_2$ is $O_i^{\mathcal{T}r_2} = \{q_1, q_2, ..., q_m\}$. If m and n are not equal, we intercept the smaller length. Additionally, we need to calculate the sum of the costs required to convert from $O_i^{\mathcal{T}r_1}$ to $O_i^{\mathcal{T}r_2}$. The formula for calculating the cost of each step is as below:

$$\delta_{i+1} = \delta_i + O_i^{\mathcal{T}r_1} - O_i^{\mathcal{T}r_2} \tag{10}$$

where δ_0 equals to 0. We can get final total cost, i.e. the *Wasserstein* distance is $W = \sum |\delta_i|$.

We apply this distance measurement to calculate the discrete trajectory point offsets. Different from *KL* divergence [9], *Wasserstein* distance is more suitable for discretely distributed trajectory points on a road. When two distributions are located in lower dimensional manifolds without overlaps, *Wasserstein* distance can still provide a meaningful and smooth representation of the distance in-between. We refer to the calculated distance as the variation degree $\mathcal{VD}$. If $\mathcal{VD}$ is greater than the offset outlier threshold th_o, this road is identified as a closed road caused by merging lanes.

In the choice of offset outlier threshold th_o, we use a combination of manual tagging and website searching to test. Consider that the road closure caused by merging lanes usually occur in the wide two-way lanes, the majority of normal driving is included when the threshold is small (i.e. the $Wasserstein$ distance is small). The best performance can be obtained when the threshold is set to 11, which we set as the offset outlier threshold, as illustrated in Figure 14(b).

5.6 Ablation Experiment

Finally, we conduct an ablation experiment, in which we use three features at the road closure detection: turning volume of strongly correlated road, U-turn frequency and trajectory offset. As shown in Table 1, it can be seen that $F1\text{-}score$ is continuously improved as the features are added, which further demonstrates the effectiveness of the experiment.

turning volume	U-turn	trajectory offset	F1-score
√			0.69
√	√		0.88
√	√	√	0.94

Table 1: Ablation Experiment

6 CONCLUSION

In this paper, we propose a *RCDM* framework to detect road closures based on multiple features. *RCDM* is composed of an *off-line closure feature modeling* phase and an *online road closure detection* phase. In order to pinpoint the road closures efficiently, we design a strategy of closed grid candidate identifying first and then road closure detecting. Additionally, to improve the detection precision, deep learning technique is leveraged to predict the values of road closure features in online detection phase. The combined model consisting of CNN and $LSTM$ and the combined model consisting of GCN and $LSTM$ are put forward to capture the spatial-temporal dependency of traffic situation from grid level and road level respectively. Experiments on real data sets demonstrate the effectiveness and efficiency of *RCDM*. In the future, we will further tackle the issue of semantic annotation for road closure events, to ensure high quality of navigation service.

ACKNOWLEDGEMENTS

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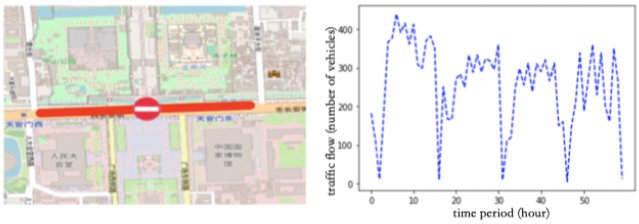
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APPENDICES

Symbols	Description
$\mathcal{V}$	the vertex includes all roads
$\mathcal{E}$	the edge represents road's reachability relationship
$\mathcal{M}$	the road map is represented as a graph
$\mathcal{T}_r$	the trajectory
tp	the position point
$\mathcal{G}$	the grid
$\mathcal{GC}$	the closed grid candidate
$\mathcal{DA}$	the detection area
$\mathcal{M}(da)$	the road map of the detection area
$\mathcal{CS}$	the correlation strength between successive roads
$\mathcal{VC}$	the closed road
$\mathcal{TF}_h$	the hour-level traffic flow of each grid
$\mathcal{RU}_h$	the hour-level U-turn frequency of each road
$\mathcal{TV}_h$	the hour-level traffic turning volume of each road
$\mathcal{S}$	the road sequence as matched result

Table 2: Summary of notations



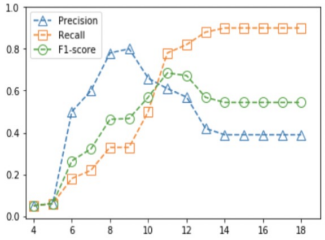
(a) Closure of East Changan Avenue (b) Trend of traffic flow



(c) Closure caused by train

Figure 13: Periodic closure





(a) Closure caused by merging lanes (b) Parameters evaluation of th_o

Figure 14: Closure caused by merging lanes and parameters evaluation